Continuous spin in Galilei mechanics: a transverse boost moment

Result, 2026-10-02. The Galilei continuous-spin analogue has a classical interpretation as a fixed transverse **boost moment**. At zero central mass and nonzero momentum, its dimensionful parameter is

$$\beta = \left|\mathbf{K} - \frac{\mathbf{K}\cdot\mathbf{P}}{|\mathbf{P}|^2}\mathbf{P}\right| = \frac{|\mathbf{P}\times\mathbf{K}|}{|\mathbf{P}|} > 0. \tag{1}$$

For Newtonian constituents, the boost charge comes from the mass-weighted position, $\sum_i m_i\mathbf{x}_i - t\mathbf{P}$. A nonempty system of positive masses cannot have zero total mass. Allowing masses $+\mu$ and $-\mu$ gives an explicit classical realization in which $\beta = \mu|\mathbf{d}_\perp|$, with $\mathbf{d}$ their relative separation (§4). Its energy is unbounded below. The elementary orbit's usual spacetime action also imposes fixed physical time (§5); the signed pair realizes charges, not an ordinary one-body trajectory on that orbit.

The orbit and representation classification is established literature. The calculations below explain its mechanical meaning; they make no novelty claim and have not been independently refereed. This is a modern Galilei-group interpretation, not a premise attributed to Newton. Section 7 tests the proposed connection between a Planck gap and a mass gap: a supplied action constant and a supplied spectral gap enter through their product in a boost uncertainty bound. A positive continuous-spin label alone gives neither gap.

1. Charges, conventions and the invariant

Work in three spatial dimensions with a Hamiltonian action of the Bargmann central extension on a free or isolated system. Applied external forces require including their sources before asserting these symmetries. Write M for central mass, H for energy, $\mathbf{P}$ for momentum, $\mathbf{J}$ for angular momentum and $\mathbf{K}$ for the boost generator at reference time $t = 0$. Use the classical Poisson conventions

$$\begin{aligned} &\{J_i, J_j\} = \epsilon_{ijk}J_k, \quad \{J_i, P_j\} = \epsilon_{ijk}P_k, \quad \{J_i, K_j\} = \epsilon_{ijk}K_k, \\ &\{K_i, P_j\} = M\delta_{ij}, \quad \{K_i, H\} = P_i, \qquad \{K_i, K_j\} = 0. \end{aligned} \tag{2}$$

M is central, $\{P_i, P_j\} = \{P_i, H\} = \{J_i, H\} = 0$, and evolution is $\dot{A} = \{A, H\}$. Along a trajectory, $\mathbf{K} - t\mathbf{P}$ is the conserved boost charge. Subtracting $t\mathbf{P}$ leaves (1) unchanged. These brackets specify our signs rather than importing signs from a coadjoint-coordinate convention.

Define

$$\mathbf{W} = M\mathbf{J} - \mathbf{K}\times\mathbf{P}, \qquad C = |\mathbf{W}|^2. \tag{3}$$

Proposition 1. *C is invariant under the connected Bargmann group. On the stratum $M = 0$, $|\mathbf{P}|^2$ is invariant as well. Thus (1) is invariant when $M = 0$ and $|\mathbf{P}| = p_0 > 0$.*

Proof. A spatial translation generated by $\mathbf{a}\cdot\mathbf{P}$ changes $\mathbf{K}$ by $M\mathbf{a}$ and $\mathbf{J}$ by $\mathbf{a}\times\mathbf{P}$, so both changes cancel in (3). A passive boost generated by $\mathbf{v}\cdot\mathbf{K}$ gives

$$\mathbf{P}' = \mathbf{P} - M\mathbf{v}, \qquad \mathbf{K}' = \mathbf{K}, \qquad \mathbf{J}' = \mathbf{J} + \mathbf{v}\times\mathbf{K}. \tag{4}$$

The two boost contributions to $\mathbf{W}'$ cancel since $\mathbf{v}\times\mathbf{K} = -\mathbf{K}\times\mathbf{v}$. Time evolution adds $t\mathbf{P}$ to $\mathbf{K}$ and preserves $\mathbf{P}, \mathbf{J}$; rotations rotate $\mathbf{W}$. Hence its norm is invariant. At $M = 0$, boosts also preserve $\mathbf{P}$, while rotations preserve its norm. Finally, $C = p_0^2\beta^2$ on this stratum. □

The massless orbit with labels $p_0 > 0, \beta > 0$ is orbit #7 in Figueroa-O'Farrill et al. (passage-read: arXiv v1, §7.2.2 and Table 1, p. 8). Its corresponding representation families are $V_\pm(p_0, \beta)$ in units $\hbar = 1$ (§11.4.5 and Table 6). At $\beta = 0$ the little-group boosts act trivially; that is a different representation class.

2. What ordinary Newtonian spin measures

For canonical Newtonian constituents at the reference time, assume

$$\{x_{ia}, p_{jb}\} = \delta_{ij}\delta_{ab}, \qquad M = \sum_i m_i > 0, \qquad \mathbf{P} = \sum_i \mathbf{p}_i, \qquad \mathbf{K} = \sum_i m_i \mathbf{x}_i.$$

Put $\mathbf{X} = \mathbf{K}/M$. If the constituents carry internal angular momenta $\mathbf{s}_i$, the mechanical total is

$$\mathbf{J} = \sum_i \mathbf{x}_i \times \mathbf{p}_i + \sum_i \mathbf{s}_i.$$

Equation (3) then becomes

$$\mathbf{W} = M\mathbf{S}, \qquad \mathbf{S} = \mathbf{J} - \mathbf{X} \times \mathbf{P} = \sum_i (\mathbf{x}_i - \mathbf{X}) \times \mathbf{p}_i + \sum_i \mathbf{s}_i. \tag{5}$$

$\mathbf{S}$ is angular momentum about the centre of mass, including internal spin. A structureless single point has $\mathbf{S} = 0$; a rotating composite can have nonzero $\mathbf{S}$. Its **classical** magnitude can vary continuously. Discrete integer or half-integer spin labels enter when one chooses unitary representations of the massive little group $SU(2)$; they are not a restriction on classical initial angular momentum.

The zero-mass parameter in (1) measures a different charge. At $M = 0$, $\mathbf{J}$ drops out of (3), and there is no division by M defining a centre of mass. In particular, β is not the spin magnitude of a massive Newtonian body.

3. The internal cylinder and the continuous label

Choose reference energy $E = 0$ and momentum $\mathbf{P} = p_0\mathbf{e}_3$. The boost laws (4) and (13) show that transverse boosts preserve this pair when $M = 0$ and $v_3 = 0$. Together with rotations about $\mathbf{e}_3$, their little-group algebra is $\mathfrak{iso}(2)$.

A classical realization of its nontrivial internal orbit uses an angle $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ and its conjugate angular momentum $\ell \in \mathbb{R}$, with $\{\theta, \ell\} = 1$:

$$K_1 = \beta\cos\theta, \qquad K_2 = \beta\sin\theta, \qquad J_3 = \ell. \tag{6}$$

Direct differentiation gives $\{J_3, K_1\} = K_2$, $\{J_3, K_2\} = -K_1$ and $\{K_1, K_2\} = 0$. The phase space is a cylinder; β fixes the radius of the boost moment, while ℓ remains an unrestricted classical coordinate. This is the internal little-group orbit, not the full Galilei particle phase space.

With a **supplied** quantum action constant, one can represent $J_3 = -i\hbar\partial_\theta$ and K_1, K_2 by multiplication by (6). Periodic and antiperiodic wave functions give respectively $\ell = \hbar n$ and $\ell = \hbar(n + \frac{1}{2})$, $n \in \mathbb{Z}$. Multiplication by $K_1 \pm iK_2 = \beta e^{\pm i\theta}$ shifts this angular-momentum label. Thus the tower is discrete although β is continuous. This construction also explains the two families listed in Figueroa-O'Farrill et al. (passage-read: §11.3 and Table 6, p. 21).

Restoring units, a boost acts through phases involving $\mathbf{v}_\perp \cdot \mathbf{K}_\perp/\hbar$. Consequently

$$[\beta] = \text{mass} \times \text{length}, \qquad [p_0\beta] = \text{mass} \times \text{action}, \qquad [\beta/p_0] = \text{time}. \tag{7}$$

In the phase convention $\exp(i\mathbf{v} \cdot \mathbf{k})$, the generator is $\mathbf{k} = \mathbf{K}/\hbar$, and the radius $\beta/\hbar$ has units of inverse velocity. Neither the angle's periodicity nor the positivity of this particular orbit label fixes $\hbar$, a universal action floor, or a dynamical period. No Hamiltonian for the internal cylinder has been assumed.

4. An explicit signed-mass realization

Proposition 2. Two free Newtonian coordinates with inertial masses $+\mu$ and $-\mu$, $\mu > 0$, realize (2) at $M = 0$ and admit arbitrary $p_0 > 0, \beta > 0$ in (1). Their Hamiltonian has no lower bound.

Proof. Allow the signed kinetic Lagrangian

$$L = \frac{\mu}{2}|\dot{\mathbf{x}}_+|^2 - \frac{\mu}{2}|\dot{\mathbf{x}}_-|^2.$$

Define $\mathbf{R} = (\mathbf{x}_+ + \mathbf{x}_-)/2$ and $\mathbf{d} = \mathbf{x}_+ - \mathbf{x}_-$. Then

$$L = \mu\dot{\mathbf{R}} \cdot \dot{\mathbf{d}}, \qquad \mathbf{P} = \mu\dot{\mathbf{d}}, \qquad \boldsymbol{\pi} = \mu\dot{\mathbf{R}}, \qquad H = \frac{\mathbf{P} \cdot \boldsymbol{\pi}}{\mu}. \tag{8}$$

Use canonical pairs $(\mathbf{R}, \mathbf{P})$ and $(\mathbf{d}, \boldsymbol{\pi})$. At reference time the charges are

$$\mathbf{K} = \mu\mathbf{d}, \qquad \mathbf{J} = \mathbf{R} \times \mathbf{P} + \mathbf{d} \times \boldsymbol{\pi}, \qquad M = 0. \tag{9}$$

The canonical brackets give $\{K_i, P_j\} = 0$, $\{K_i, H\} = P_i$, $\{K_i, K_j\} = 0$; the two orbital angular momenta in (9) generate the required rotations. Also $\{\mathbf{J}, H\} = 0$ and $\{\mathbf{P}, H\} = 0$. Thus all of (2) holds. Along the free motion,

$$\mathbf{d}(t) = \mathbf{d}(0) + \frac{t}{\mu}\mathbf{P}, \qquad \mathbf{K}_{\text{conserved}} = \mu\mathbf{d}(t) - t\mathbf{P}, \qquad \beta = \mu|\mathbf{d}_\perp(0)|. \tag{10}$$

For example, $\mathbf{P} = p_0\mathbf{e}_3$ and $\mathbf{d}(0) = (\beta/\mu)\mathbf{e}_1$ give any desired positive labels. Keeping these fixed while varying π_3 makes $H = p_0\pi_3/\mu$ range over $\mathbb{R}$. □

The vector $\mu\mathbf{d}$ is a mass dipole: translating both bodies leaves it unchanged because their masses sum to zero. Its transverse part also survives a shift of the time origin, which adds only a multiple of $\mathbf{P}$. This gives (1) a concrete mechanical meaning. For nonnegative constituent masses, zero total mass would instead force every mass to vanish; ordinary positive-mass matter cannot supply this dipole.

The signed pair's twelve-dimensional phase space contains a family of charge orbits. Equations (8)–(10) are a realization of the algebra and its labels, not an identification of that phase space with one eight-dimensional coadjoint orbit. Its ordinary time evolution must not be transferred to the elementary particle interpretation below.

5. The elementary spacetime orbit has fixed time

For the elementary orbit, take fixed perpendicular vectors $\mathbf{p}_*, \mathbf{b}_*$ of lengths p_0, β and put $\mathbf{P} = R\mathbf{p}_*$, $\mathbf{B} = R\mathbf{b}_*$, $R \in SO(3)$. The standard spacetime orbit action is

$$S = \int \left[\mathbf{P} \cdot \mathbf{x}' + \mathbf{B} \cdot \mathbf{v}' + (\mathbf{P} \cdot \mathbf{v})t'\right] d\tau, \qquad \mathbf{P} \cdot \mathbf{B} = 0. \tag{11}$$

Primes denote the arbitrary curve parameter τ; t is physical time and $\mathbf{v}$ is an auxiliary boost coordinate. We reproduce Figueroa-O'Farrill et al. (passage-read: arXiv v1, §8.3.5, equations (8.51)–(8.60), p. 15). The coefficient $\mathbf{B}$ is conjugate to this velocity coordinate, not a canonical momentum conjugate to $\mathbf{x}$.

Varying $\mathbf{x}$ and $\mathbf{v}$ yields $\mathbf{P}' = 0$ and $\mathbf{B}' = \mathbf{P}t'$. Differentiating the identity $\mathbf{P} \cdot \mathbf{B} = 0$ then gives

$$0 = \mathbf{P} \cdot \mathbf{B}' = p_0^2 t', \qquad t' = 0. \tag{12}$$

Thus these curves relate simultaneous events. They are not worldlines $\mathbf{x}(t)$ of a freely moving massive Newtonian body. Time translations still act on the orbit and its representations; (12) does not say that their generator is zero.

Another obstruction to an ordinary point-particle phase space follows without using (11). On a fixed orbit $|\mathbf{P}|^2 = p_0^2$ is a constant function. Three position observables with $\{x_i, P_j\} = \delta_{ij}$ would imply $0 = \{x_i, |\mathbf{P}|^2\} = 2P_i$, contradicting $p_0 > 0$. The elementary orbit therefore cannot retain all three canonical positions from the unreduced signed pair.

6. Energy positivity excludes the whole moving-momentum sector

The passive boost law follows by integrating (2):

$$H' = H - \mathbf{v} \cdot \mathbf{P} + \frac{M}{2}|\mathbf{v}|^2. \tag{13}$$

At $M = 0$, choosing $\mathbf{v} = \lambda \mathbf{P}$ gives $H' = H - \lambda p_0^2$. Hence the classical energy takes all real values on every orbit with $p_0 > 0$. In the induced unitary representation the energy-momentum support is likewise $\mathbb{R} \times S^2_{p_0}$, so the energy operator has spectrum $\mathbb{R}$ (Figueroa-O'Farrill et al., passage-read: §11.1, Table 5, p. 20). The conclusion applies at $\beta = 0$ as well: the obstruction is zero central mass with nonzero momentum, not the nontrivial internal tower alone. It makes no statement about the distinct $M = 0, \mathbf{P} = 0$ sectors.

Matching little groups also does not specify a nonrelativistic limit of a Poincaré continuous-spin particle. With fixed nonzero relativistic momentum, its massless energy $E = c|\mathbf{p}|$ diverges as $c \to \infty$. A proposed contraction must state its changing labels and energy subtraction. No such contraction is constructed here.

7. Does zero mass connect a Planck gap and a mass gap?

The quantum Galilei algebra exposes one connection:

$$[\widehat{K}_i, \widehat{P}_j] = i\hbar M \delta_{ij} I, \qquad \Delta_\psi K_i \, \Delta_\psi P_i \geq \frac{\hbar |M|}{2}. \tag{14}$$

The inequality assumes a supplied $\hbar > 0$, a fixed central-mass sector and a normalized vector for which the variances and commutator expectation exist. The unscaled commutator vanishes if either factor vanishes. This does not identify their classical limits: for a massive body the rescaled bracket $[K_i, P_j]/(i\hbar)$ tends to the nonzero Poisson bracket $M\delta_{ij}$ as $\hbar \to 0$.

Neither factor in (14) is fixed by the other. Positive central mass also does not force a positive energy threshold: the free massive Galilei Hamiltonian $\mathbf{P}^2/(2M)$ on $L^2(\mathbb{R}^3)$ has spectrum $[0, \infty)$ for every $M > 0$. Excluding the exact zero-mass sector also leaves arbitrarily small positive mass labels available; exclusion of a point is weaker than a gap above that point.

At finite c, the corresponding Poincaré boost bracket uses energy:

$$[\widehat{K}_i, \widehat{P}_j] = \frac{i\hbar}{c^2} \delta_{ij} \widehat{H}. \tag{15}$$

K_i is normalized as the generator conjugate to infinitesimal velocity, with the same dimensions as the Galilei boost moment. Equation (15) is the usual Lorentz-translation bracket with physical units restored; see Brink et al. (passage-read: §2, p. 2, in units $c = \hbar = 1$).

On a massive branch, writing $H = Mc^2 + H_{\mathrm{nr}}$ gives $H/c^2 = M + H_{\mathrm{nr}}/c^2$. On state families with bounded residual-energy matrix elements, this coefficient tends to M as $c \to \infty$ and recovers (14). This specifies why a massive nonrelativistic sector has central mass; it does not equate that label with a gap of the full theory.

Proposition 3 (a supplied spectral gap gives a boost uncertainty floor). Suppose an already constructed Poincaré-covariant quantum theory has a normalized invariant vacuum Ω, physical Hamiltonian $H\Omega = 0$, and $H \geq \Delta I$ on $\Omega^\perp$, with $\Delta > 0$ in energy units. Assume self-adjoint boost and momentum generators satisfying (15) on a common invariant domain where the commutator expectation is defined. Every normalized $\psi \perp \Omega$ in that domain with finite variances satisfies

$$\Delta_\psi K_i \, \Delta_\psi P_i \geq \frac{\hbar \langle H \rangle_\psi}{2c^2} \geq \frac{\hbar \Delta}{2c^2}. \tag{16}$$

Proof. Cauchy–Schwarz applied to the centred vectors $(K_i - \langle K_i \rangle)\psi$ and $(P_i - \langle P_i \rangle)\psi$ bounds the modulus of their inner product by $\Delta_\psi K_i \Delta_\psi P_i$. Its imaginary part has modulus $\frac{1}{2}|\langle [K_i, P_i] \rangle|$. Use (15) and the assumed spectral bound. □

The mass-equivalent threshold is Δ/c^2, so the right side is half the product of that threshold and the supplied action constant. Its units are **mass times action**. It is a floor for a boost–momentum uncertainty product, not for Newton's inertial–parabola comparison. The proposition assumes the physical spectral gap after the required volume and continuum limits; it constructs neither that theory nor those limits, and imposes no lower bound on which positive value of $\hbar$ may be supplied. The converse cannot be obtained by reversing (16): a lower bound on an uncertainty product does not, through this inequality, give a lower bound on energy.

A positive continuous-spin parameter gives a direct counterexample to identifying every positive representation scale with a mass gap. Poincaré continuous-spin representations are massless despite their nonzero Pauli–Lubanski invariant (Brink et al., passage-read: §§2–2.1, pp. 3–4). On their positive-energy light cone, a longitudinal boost changes E to $e^{-\eta}E$ without changing the continuous-spin label. As $\eta \to \infty$, $E \to 0$. Such a nonvacuum sector therefore has no positive energy gap, even with $\hbar > 0$ and the internal label fixed away from zero.

Zero mass is thus a useful test of which positivity premise a comparison needs. A mechanism generating a physical gap must also control access to arbitrarily small nonvacuum energies. In the solved matrix model, transverse zero-point confinement provides such a dynamical mechanism; the group classification alone provides none. See the action-floor/gap note (passage-read: §§3–5 and the 2026-10-02 correction): in four-dimensional Yang–Mills an action unit is already available from the classical coupling, while a generated energy scale remains a separate requirement.

References

Figueroa-O'Farrill, José Miguel; Pekar, Simon; Pérez, Alfredo; and Prohazka, Stefan. *Galilei particles revisited.* SciPost Physics Lecture Notes **93** (2025), DOI 10.21468/SciPostPhysLectNotes.93. Text read: arXiv:2401.03241v1 (2024), selected passages specified above; not a full-paper proof audit. Bibliographic metadata checked against arXiv, INSPIRE and Crossref on 2026-10-02. The published article and the read preprint have different pagination; all page and equation locators here refer to the preprint.

Brink, Lars; Khan, Abu M.; Ramond, Pierre; and Xiong, Xiaozhen. *Continuous spin representations of the Poincaré and super-Poincaré groups.* Journal of Mathematical Physics **43** (12), 6279–6295 (2002), DOI 10.1063/1.1518138. Text read: arXiv:hep-th/0205145v1, §§2–2.1, especially preprint pp. 2–4. Bibliographic metadata checked against arXiv, INSPIRE and Crossref on 2026-10-02.

8. Consequence for STATE

Use this comparison when interpreting group labels as mechanical quantities: classical massive spin, a massless transverse boost moment, and a supplied quantum action constant are separate inputs. Equation (16) records a conditional connection to a physical energy gap; it supplies no positive-mass completion of Newton's mechanics, no selection of universal action, and no Yang–Mills construction. Continue the Newton necessity queue in STATE, retaining the common-scale versus positive-scale distinction in rotation composition (passage-read: opening correction and §§1–2). Any use of zero mass as a new mechanism must specify the surviving observables and the physical Hamiltonian instead of transferring a positive orbit label into a spectral claim.